

Hans Kersting, PhD

ML Research Scientist — Programmatic Advertising

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ML scientist with 10+ years in research and production, specializing in real-time bidding and budget pacing. Track record shipping models at scale, leading cross-functional teams, and designing experiments to measure customer impact.

Skills

Programming **Python, SQL, Pytorch, Tensorflow**, Spark, Hadoop, AWS, Git
Expertise **Machine Learning, Real-Time Bidding, Budget Pacing, Control, Experiment Design**, Deep Learning, Probability Theory, Bayesian Methods, Optimization
Languages Fluent **English** and **German**

Experience

- 01/2023–present **Research Scientist**, *Yahoo*, Berlin, Germany
ML R&D for Yahoo's demand-side platform (DSP), serving **500B+ bid requests/day** at millions of QPS.
- **Led cross-functional team** (4 engineers, 2 data scientists) to **rapidly ship** new ML product feature end-to-end—from design docs to **production**
 - Designed new **click prediction model** (field-weighted factorization machine), now standard across DSP: **+5.6% lower CPC**, **+3.5% CTR**, **+8% fees**
 - Designed and shipped a **delivery acceleration** feature: **+1.3% spend** across the DSP
 - Co-developed novel **online SGD algorithm** for real-time bid optimization; deployed to production (runs every 5 mins)
 - Partnered with **product managers** on feasibility and timelines; designed experiments to measure customer impact
 - Senior technical point of contact for the control algorithm—**mentored engineers** and advised cross-team investigations
- 10/2020–12/2022 **Postdoctoral Researcher**, *INRIA & ENS Paris*, France
Research on **optimizers for deep learning**. Published at **ICML**, **NeurIPS**, and **AISTATS**. Co-organized reading groups and workshops.
- 07/2019–09/2019 **Research Intern**, *Amazon Research*, Cambridge, UK
Research on **ML for supply-chain optimization**.
- 10/2017–02/2018 **Research Intern**, *Bosch Center for AI (BCAI)*, Stuttgart, Germany
Research on **inverse problems** for engine control.

Education

- 2015–2020 **PhD in Machine Learning**, *Max Planck Institute & Univ. Tübingen*
Grade: **magna cum laude**. Thesis on probabilistic numerics and Bayesian inference.
- 2013–2015 **MSc in Mathematics**, *LMU Munich*, GPA: 1.44 (very good)
Specialization in stochastic calculus and probabilistic modelling.
- 2009–2013 **BSc in Mathematics**, *LMU Munich*, GPA: 1.14 (very good)
Specialization in probability theory.

Selected Publications

Book **Probabilistic Numerics – Computation as Machine Learning**, Hennig, Osborne, **Kersting**. Cambridge University Press, 2022.

- ICML 2022 Anticorrelated Noise Injection for Improved Generalization. Orvieto*, **Kersting***, Proske, Bach, Lucchi.
- NeurIPS 2022 On the Theoretical Properties of Noise Correlation in Stochastic Optimization. Lucchi*, Proske*, Orvieto, Bach, **Kersting**.
- ICML 2020 Differentiable Likelihoods for Fast Inversion of 'Likelihood-Free' Dynamical Systems. **Kersting***, Krämer*, et al.

Full list: Google Scholar (h-index: 11, 660+ citations)

Recognition

- 2022 **Outstanding Reviewer Award, ICML**
Top 10% of reviewers.
- 2013–2015 **German Academic Scholarship Foundation, Studienstiftung**
Awarded to <0.5% of German university students.